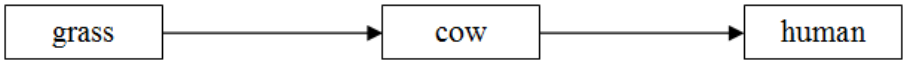
Structured questions

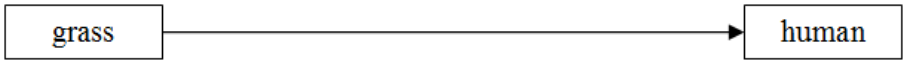
|!|BCELB0200300001|!| ★

The diagram below shows two agricultural food chains.

Food chain 1



Food chain 2



The table below shows the annual energy yield for human consumption of the two food chains.

	Annual energy yield for human consumption (kJ per m^2 per year)
Food chain 1	124.5
Food chain 2	1080

(a) Much of the energy captured by the grass does not transfer to the cows in Food chain 1. Give ***two*** alternative destinations for this energy. (2 marks)

(b) Which organism in Food chain 1 is a secondary consumer? (1 mark)

(c) Explain why Food chain 2 provides a higher energy yield for human consumption than Food chain 1. (2 marks)

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(d) Some farmers keep cows outdoors, while others keep cows indoors. Suggest why keeping cows indoors leads to more efficient growth. (2 marks)

##

(a) There is energy loss as heat / excretory products / trapped in the dead bodies of uneaten grass. (1)
There may be other herbivores competing for the grass. (1)
(b) Human (1)
(c) Humans are in a lower trophic level in Food chain 2 than in Food chain 1. (1)
Less energy is lost in the transfer between trophic levels. (1)
(d) When cows are kept indoors, they spent less energy in movement. (1)
They are kept warm, so less energy is lost as heat. (1)

(7 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Analysing data	1
(c)	Applying concepts	2
(d)	Applying concepts	2

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|!|BCELB0200800002|!| ★ **SI**

Researchers conducted an investigation to determine whether the parasitic dodder weed can act as a biological control to the invasive weed *Mikania micrantha*. Sampling was done on two plots of 50 m^2 grassland areas where *M. micrantha* was abundant. Plot 1 contained *M. micrantha* only while Plot 2 contained both *M. micrantha* and dodder, the latter of which was introduced two months prior sampling. The data collected from the field study is shown in the table below.

Random sampling point	**Abundance of *M. micrantha* (Arbitrary units)**	
	Plot 1 (*M. micrantha* only)	Plot 2 (*M. micrantha* and dodder)
1	75	44
2	60	55
3	83	31
4	91	49
5	74	28
6	59	34
7	66	35
8	53	26
	Mean Abundance (Arbitrary units)	
	70.13	?

(a) Suggest a sampling method that can produce the data above and briefly describe how it should be carried out. As part of your answer, choose an appropriate method for describing the abundance of *M. micrantha*. Justify your choice. (5 marks)

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(b) Calculate the mean abundance for Plot 2. Show your working. (2 marks)

(c) What potential conclusion can be drawn from the data? (1 mark)

(d) Although the dodder is parasitic in nature, the tabulated data above is insufficient to show that *M. micrantha* acts as a host.

(i) What are the criteria that determine whether the relationship between the dodder and *M. micrantha* is parasitism? (2 marks)

(ii) Suggest what further investigation can be done to accurately categorize this ecological relationship. (1 mark)

##

(a) Random sampling could have been used for this investigation. (1)
For each plot, quadrats are placed randomly at different locations within the study area and abundance is measured within each quadrat. (1)
Each quadrat's abundance values are then used to calculate the mean values for each plot. (1)
Percentage cover should be used to measure abundance (1)
because vines are difficult to count individually. (1)

(b) 37.75 (1)
(44 + 55 + 31 + 49 + 28 + 34 + 35 + 26) / 8 = 37.75 (1)

(c) The abundance of *M. micrantha* decreases in the presence of dodder. (1)

(d) (i) The two species must be closely associated with each other. (1)
The dodder would gain benefit from *M. micrantha*, while *M. micrantha* would be harmed by the dodder. (1)

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(ii) Any ***one*** of the following: (1)

- Measure the change in abundances of *M. micrantha* and dodder simultaneously.
- Check for close associations between the two species.
- Observe whether the dodder obtains nutrients from *M. micrantha*.

(Or other reasonable answers)

(11 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	5
(b)	Mathematical skills	2
(c)	Analysing data	1
(d)(i)	Understanding basic concepts	2
(d)(ii)	Applying concepts	1

|!|BCELB0200200003|!| ★

A lava flow from a volcanic eruption destroys a woodland habitat and hardens into volcanic rock. The table below shows the subsequent change in abundance for four types of plant in the next 100 years.

Plant type	Relative Abundance (%)		
	5 years after eruption	50 years after eruption	100 years after eruption
A	20	65	10
B	80	20	5
C	0	10	75
D	0	5	10

(a) Name the ecological process that accounts for the above changes after the volcanic eruption. Justify your answer. (3 marks)

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(b) Suggest what kinds of organism plant type B and D could be. (2 marks)

(c) In 1980, a volcanic eruption occurred at Mount St Helens in Washington, USA. Nitrogen-fixing herbaceous plants like lupins subsequently colonized the barren volcanic rock in the surrounding area as ecological succession occurred.

(i) Why are lupins particularly vital for this case of succession? (2 mark)

(ii) Herbivores such as caterpillars consume lupins in the area and can both facilitate and inhibit succession. Why? (2 mark)

##

(a) The process that occurs after the volcanic eruption is primary succession. (1)
This is because the species composition of the habitat changes (1)
over a relatively long period of time in a barren area. (1)

(b) B: lichens / mosses (1)
D: trees (1)

(c) (i) The lupins can survive on the nitrogen-deficient volcanic rock because they can convert nitrogen from the air into ammonium compounds. (1)
The volcanic soil is also enriched with ammonium compounds as they are released from the dead remains of nitrogen-fixing bacteria in the lupins. (1)

(ii) Herbivores can enrich the soil by leaving behind organic waste and dead remains. (1)
Herbivores consume plants and directly reduce their abundance, thus slowing down the rate at which the plants can recolonize and modify the area. (1)

(9 marks)

##

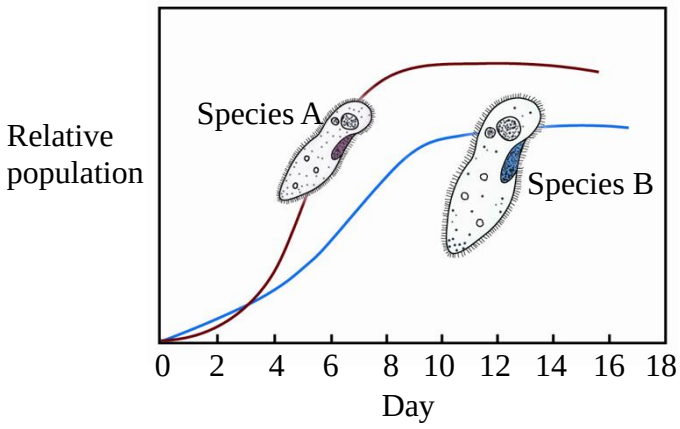
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	Skills assessed	Marks
(a)	Applying concepts	3
(b)	Analysing data	2
(c)(i)	Applying concepts	2
(c)(ii)	Applying concepts	2

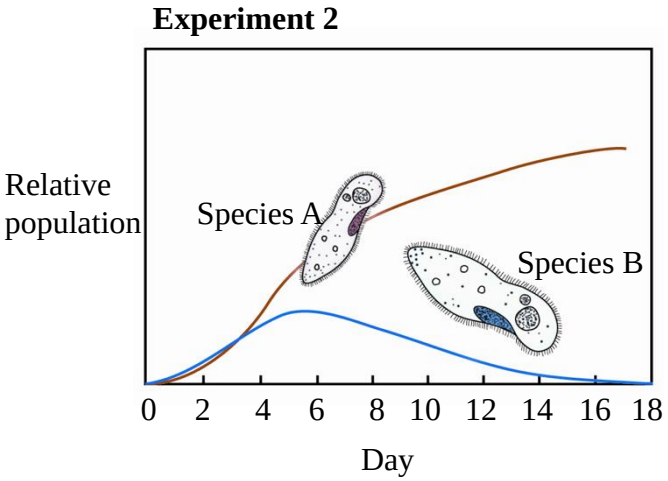
|!|BCELB0200200004|!| ★ **SI** **Cross topic**

A study was carried out to investigate the growth of two species of *Paramecium* (A and B). In experiment 1, both species were first grown under identical conditions in separate containers with an adequate supply of food. In experiment 2, species A and B were then grown together in a container with a limited supply of food. The changes in the relative populations of A and B are shown below.

Experiment 1



Experiment 2



(a) State the kingdom to which *Paramecium* belongs. (1 mark)

(b) Name the type of competition that occurs between species A and B. (1 mark)

(c) Suggest why the population of species A is larger than that of species B in experiment 1? (Hint: consider the size difference between species A and B as illustrated in the graphs.) (3 marks)

(d) Explain how the populations of species A and B change when they were grown together in experiment 2. (5 marks)

(e) In the natural environment, the population size of species B would rarely drop to zero even if species A was also present in the same habitat. Explain why. (1 mark)

##

(a) Protista (1)

(b) Interspecific competition (1)

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(c) The size of species B is larger than that of species A. (1)

Species B devotes more of the energy it gained from consuming food to increase its cell size. (1)

Species A devotes more of the energy it gained from consuming food to carry out cell division. (1)

(d) Initially, no competition existed between the two species as food was abundant and their populations were small. (1)

The population of both species grew as they continued to reproduce. (1)

The larger populations led to a scarcity of food / resources, and so competition occurred between the two species. (1)

The population of species B declined and was eventually eliminated at the end of the experiment. (1)

This was because species A outcompeted species B. (1)

(e) Any ***one*** of the following: (1)

- Species B can move to other habitats to reduce the competition it faces.
- Food supply is not absolutely limited in the natural environment.

(Or other reasonable answers)

(11 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Applying concepts	1
(c)	Applying concepts	3
(d)	Analysing data	5
(e)	Applying concepts	1

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|!|BCELB0200300005|!| ★

The following lists some of the organisms that inhabit a given habitat.

grass, rabbit, cow, grasshopper, sparrow, hawk, tiger, human

(a) Construct a food web for the organisms above. (2 marks)

(b) (i) What is the maximum number of trophic levels in this food web? (1 mark)

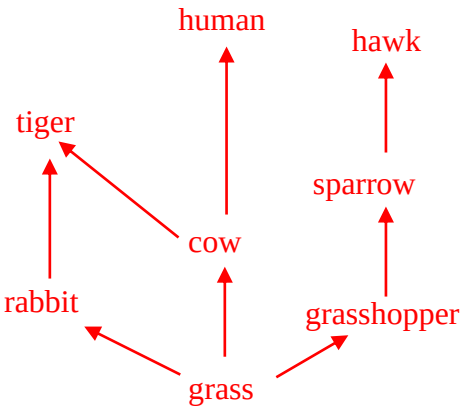
(ii) Suggest ***two*** reasons why some energy cannot be transferred from one trophic level to a higher one. (2 marks)

(c) Decomposers are not shown in the food web above. Describe their significance in the food web. (3 marks)

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##

(a)



Correct order of trophic levels (1)

Use of arrows to indicate correct feeding relationship (1)

(b) (i) 4 (1)

(ii) Any ***two*** of the following: (2)

- Not all organisms at one trophic level are eaten by organisms at the next trophic level.
- Much of the energy in organisms at one trophic level is released as heat during respiration.
- Certain food material eaten by an organism cannot be digested and absorbed.

(Or other reasonable answers)

(c) Decomposers break down the organic waste / dead remains of organisms into inorganic compounds at each trophic level. (1)

These inorganic compounds are absorbed by plants (the grass) (1)

and can thus re-enter the food web. (1)

(8 marks)

##

	Skills assessed	Marks
(a)	Presenting information	2
(b)(i)	Analysing data	1
(b)(ii)	Understanding basic concepts	2
(c)	Understanding basic concepts	3

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|!|BCELB0200300006|!| ★

The following lists some of the organisms that inhabit a woodland community.

tree, caterpillar, sparrow, hawk

(a) Which of the animals above are herbivores and carnivores, respectively? (2 marks)

(b) Draw a pyramid of numbers for the organisms above. (2 marks)

(c) Explain why the pyramid of numbers above is different from a typical one. Accordingly, identify a limitation to using pyramids of numbers. (3 marks)

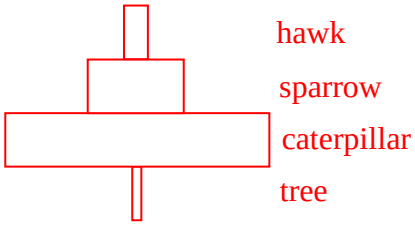
##

(a) Herbivores: caterpillar (1)

Carnivores: sparrow and hawk (1)

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(b)



Correct shape (1)

Correct labels (1)

(c) The size of the producer (tree) is very large and can support a large number of herbivores. (1)

Thus the resulting pyramid of numbers has a narrower base compared to a typical one. (1)

This highlights how pyramids of numbers fail to account for the relative body sizes of organisms at each trophic level. (1)

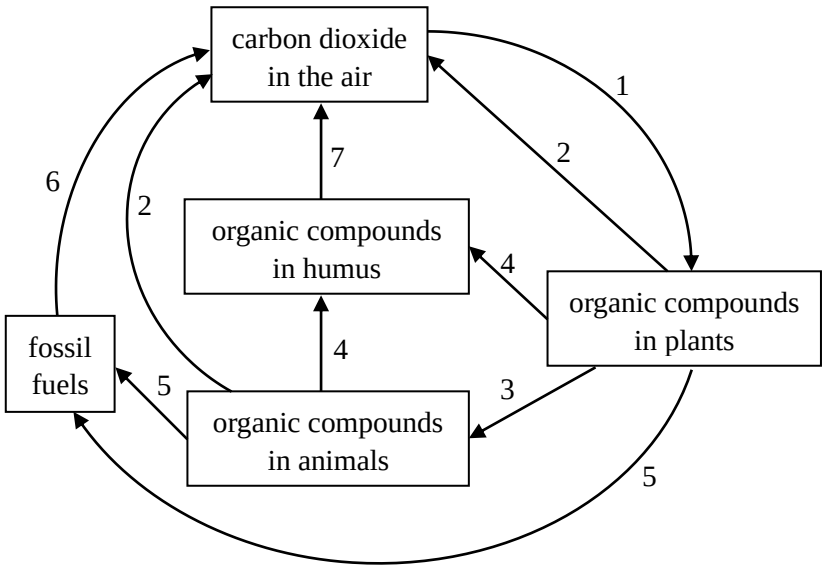
(7 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	2
(b)	Presenting information	2
(c)	Applying concepts	3

|!|BCELC0200400007|!|

The diagram below shows a simplified carbon cycle.



(a) Name processes 1 and 2. (2 marks)

(b) Which of the processes above is a human activity? State and explain ***one*** of its harmful effects. (3 marks)

(c) Describe how humus is formed during process 4. (2 marks)

##

(a) Process 1: photosynthesis (1)
Process 2: respiration (1)

(b) Process 6 (1)
Any ***one*** of the following sets: (2)

- The combustion of fossil fuels will release many air pollutants (1) that are harmful to the environment and the human body. (1)
- The combustion of fossil fuels will release greenhouse gases (1) that cause global warming. (1)

(c) Detritivores and microorganisms (e.g. bacteria, fungi) in the soil break down the organic compounds (1)
within the waste and dead remains of plants and animals, thus forming humus. (1)

(7 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Understanding basic concepts	3
(c)	Understanding basic concepts	2

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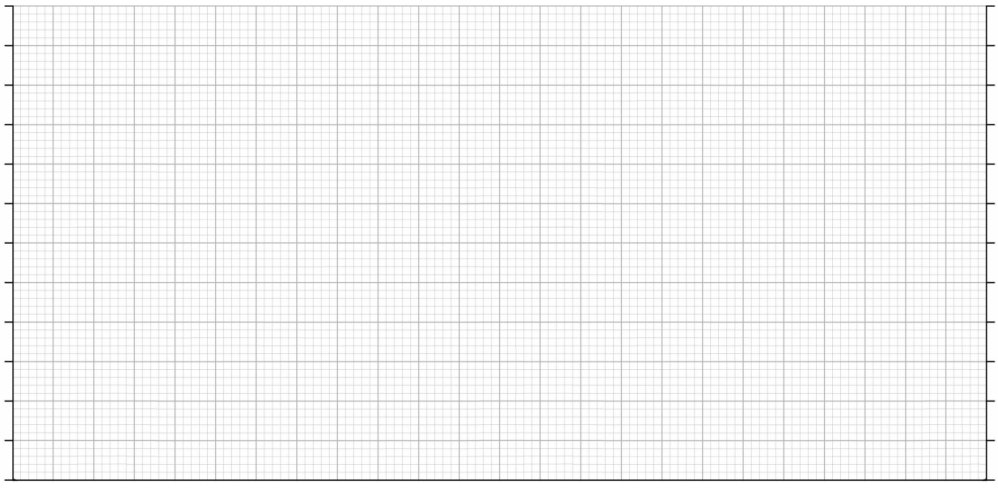
|!|BCELA0200200008|!| ★★ **SI**

In an investigation, scientists studied the relationship between moose and grey wolves in a North American national park. The following table shows the changes in the population sizes of the two species between 1959 and 2006.

Year	Population size		Year	Population size		Year	Population size	
	Moose	Grey wolves		Moose	Grey wolves		Moose	Grey wolves
1959	560	20	1975	1360	41	1991	1310	12
1960	610	22	1976	1280	44	1992	1590	12
1961	630	22	1977	1140	34	1993	1880	13
1962	640	23	1978	1000	40	1994	1770	17
1963	660	20	1979	1030	43	1995	2420	16
1964	710	26	1980	910	50	1996	1160	22
1965	740	28	1981	860	30	1997	500	24
1966	770	26	1982	870	14	1998	700	14
1967	910	22	1983	930	23	1999	750	25
1968	1040	22	1984	1040	24	2000	850	29
1969	1270	17	1985	1120	22	2001	900	19
1970	1300	18	1986	1200	20	2002	1100	17
1971	1440	20	1987	1270	16	2003	900	19
1972	1500	23	1988	1340	12	2004	750	29
1973	1430	24	1989	1400	12	2005	540	30
1974	1470	31	1990	1220	15	2006	450	30

(a) Using the data above, plot a graph below to show the changes in the population sizes of moose and grey wolves. Include an appropriate title and axis labels.

(4 marks)



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(b) According to the plotted graph, what is the ecological relationship between moose and grey wolves? (1 mark)

(c) Suggest ***two*** reasons why the population of grey wolves is smaller than that of moose. (2 marks)

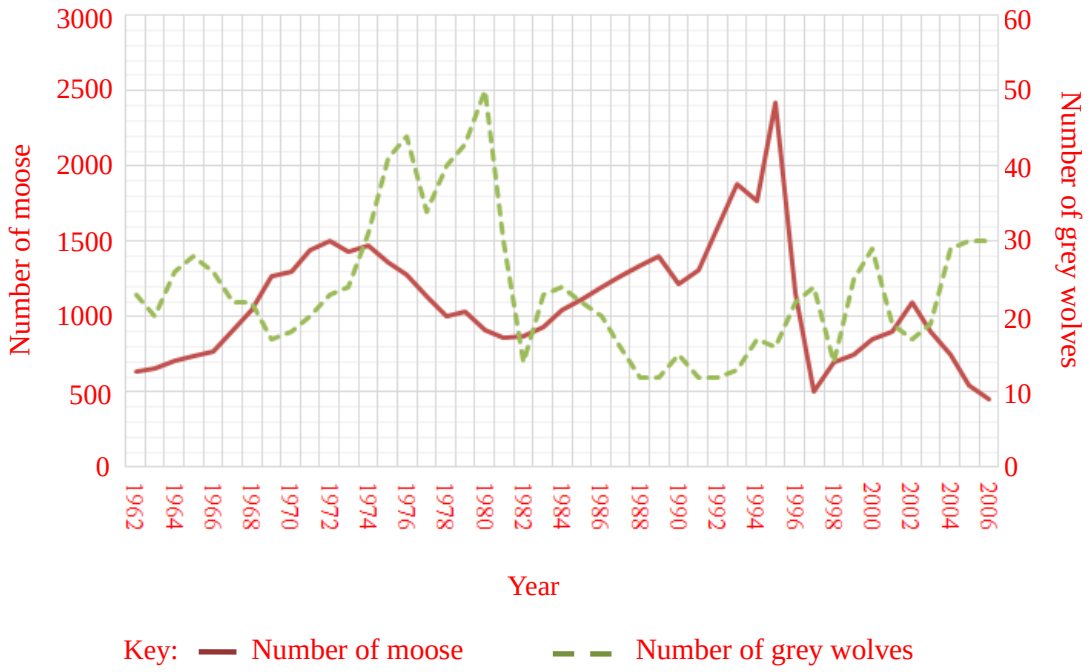
(d) Explain the shape of the two curves in the plotted graph. (5 marks)

##

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(a)

The change in population levels for moose and grey wolves between 1959–2006



Correct labelling of axis (1)
Correct plotting and joining of points (1, 1)
Correct title (1)

(b) Predation (1)

(c) Any ***two*** of the following: (2)

- The moose has a larger food supply / food supply with larger biomass.
- Energy is lost through uneaten organisms / egestion and excretion / respiration when energy is transferred from moose to grey wolves.
- Not all grey wolves can reproduce due to social hierarchies.
- The life cycle of moose is shorter than that of grey wolves.

(Or other reasonable answers)

(d) When the population of moose (the prey) increases, more food would be available to the grey wolves (the predator). (1)
This leads to an increase in the population of grey wolves. (1)
The resulting increase in predation causes the moose population to decrease. (1)
The reduced food supply cannot sustain the grey wolves, whose population decreases as a result. (1)
With fewer predators, the moose population increases again and the cycle repeats. (1)

(12 marks)

##

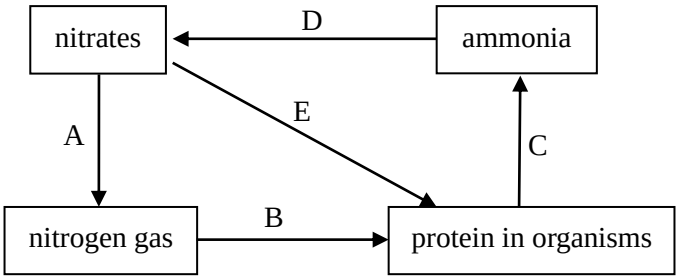
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	Skills assessed	Marks
(a)	Presenting information	4
(b)	Analysing data	1
(c)	Applying concepts	2
(d)	Applying concepts	5

|!|BCELC0200500009|!| **Cross topic**

✂

The following figure shows a simplified nitrogen cycle.



(a) Name processes A and D. (2 marks)

_____

_____

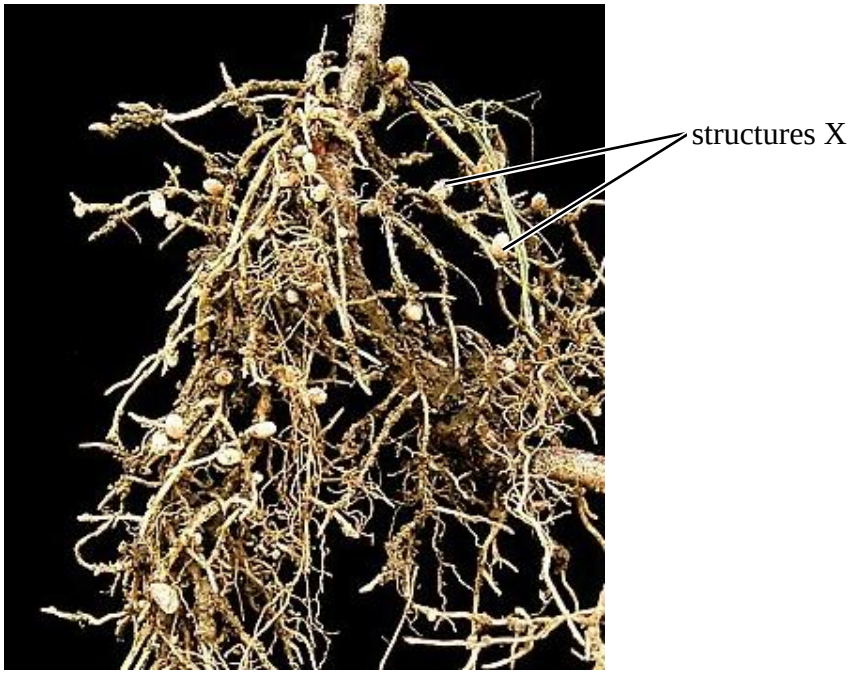
(b) What would happen to a plant if process E is inhibited? (1 mark)

_____

_____

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(c) The following photograph shows the structures of a leguminous plant where process B occurs.



(Image source: https://commons.wikimedia.org/wiki/File:Lotus_pedunculatus10_ies.jpg)

(i) Name structure X. (1 mark)

(ii) Describe how nitrogen gas can be converted into proteins in leguminous plants through process B. (3 marks)

(iii) Describe the mutualistic relationship that can be found in these structures. (2 marks)

##

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(a) A: denitrification (1)

D: nitrification (1)

(b) Yellowing of leaves / poor growth of plants (1)

(c) (i) Root nodules (1)

(ii) Nitrogen-fixing bacteria living in the root nodules of a leguminous plant can convert nitrogen gas into ammonium compounds. (1)

The leguminous plant can then obtain these ammonium compounds, which are used to synthesize amino acids. (1)

The amino acids are used to form proteins. (1)

(iii) The nitrogen-fixing bacteria in the root nodules supply fixed nitrogen to the host plant for protein synthesis. (1)

Meanwhile, the host plant provides carbohydrates for the bacteria. (1)

(9 marks)

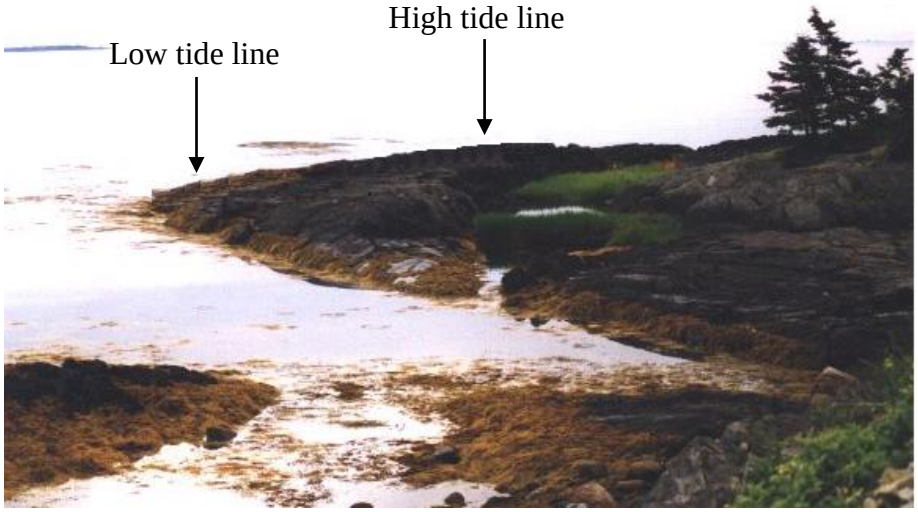
##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Applying concepts	1
(c)(i)	Analysing data	1
(c)(ii)	Understanding basic concepts	3
(c)(iii)	Applying concepts	2

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|!|BCELA0200800010|!| ★★ **SI**

During a field trip, Rose and her classmates recorded the distribution of five different species of seaweed on a rocky shore, which is shown in the photograph below.

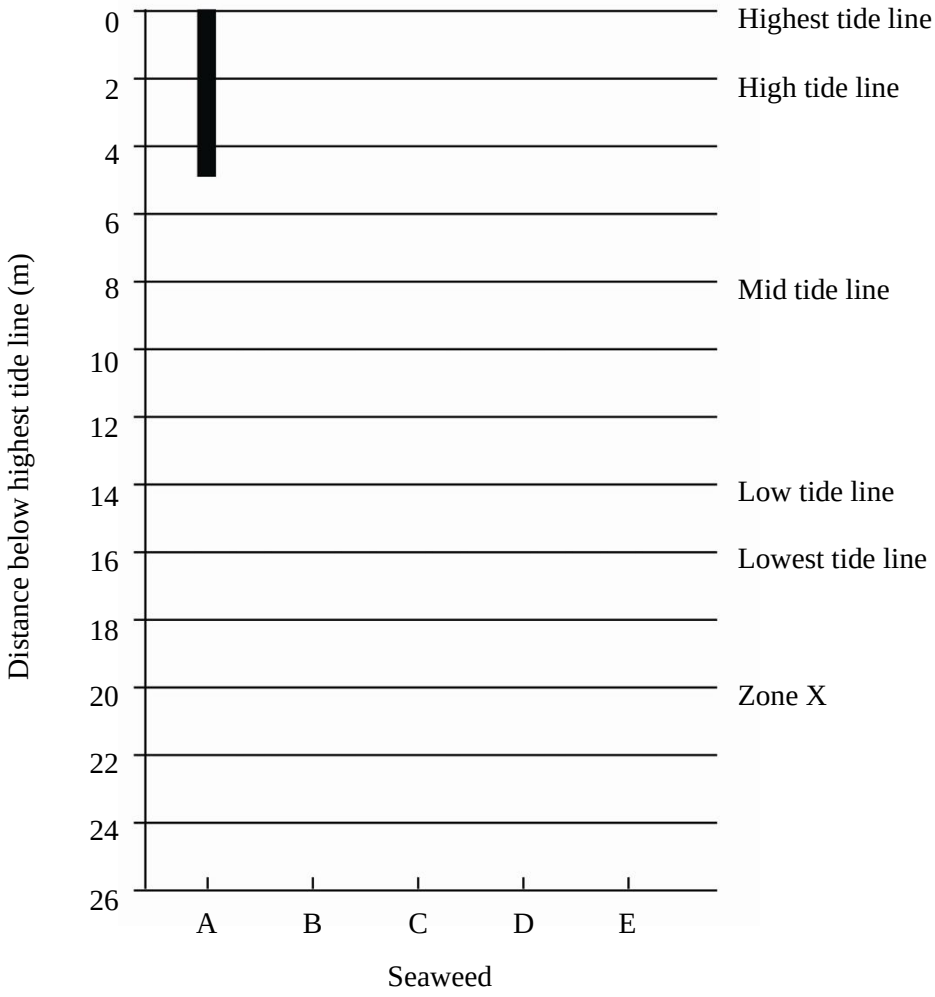


The following table shows the locations at which the different species were found.

Seaweed	Distribution
A	From the highest tide line to 3 m below the high tide line
B	From the high tide line to 2 m below the mid tide line
C	From the mid tide line to the low tide line
D	From 3 m above the low tide line to 1 m below the lowest tide line
E	From the lowest tide line to 3 m below the lowest tide line

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(a) Complete the bar graph below to show the distribution of the different species of seaweed. The bar for seaweed A is done as an example. (4 marks)



(b) Which seaweed species was found between the low tide line and the lowest tide line? (1 mark)

_____

(c) Suggest why no seaweeds were found below zone X. (2 marks)

_____

_____

(d) In this field trip, Rose and her classmates studied how the distributions of seaweeds are affected by an abiotic factor. What is the abiotic factor? (1 mark)

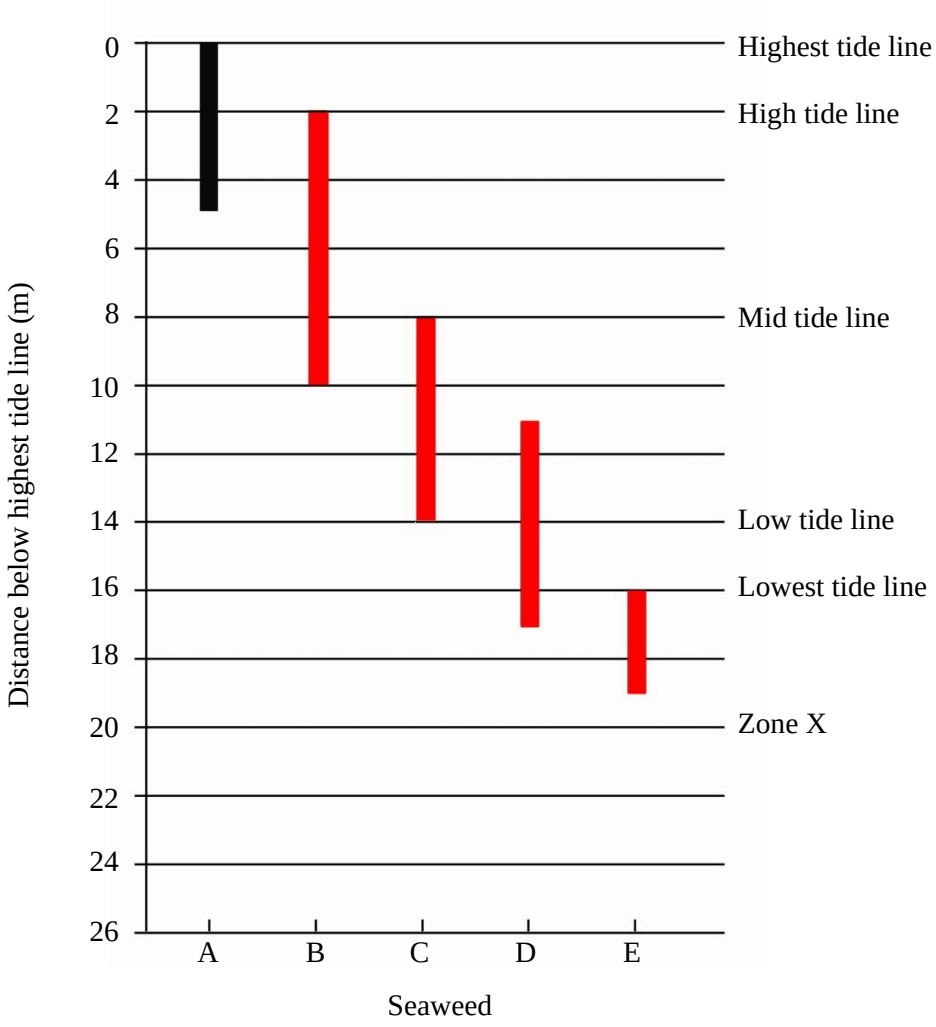
_____

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(e) Deduce which seaweed species cannot survive when exposed to the air. (2 marks)

##

(a)



Correct drawing of bars for B, C, D and E. (1 mark each) (4)

(b) D (1)

(c) Any ***one set*** of the following: (2)

- Less red and blue light can reach these areas. (1)
  The seaweeds cannot undergo photosynthesis efficiently to make food. (1)
- The areas below zone X are submerged, (1)
  making it difficult to find seaweeds in the vicinity. (1)

(Or other reasonable answers)

(d) Tides / water (1)

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(e) E (1)

The seaweed can only survive underwater because it will become desiccated if it is exposed to the air. (1)

(10 marks)

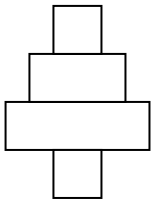
##

	Skills assessed	Marks
(a)	Presenting information	4
(b)	Analysing data	1
(c)	Applying concepts	2
(d)	Applying concepts	1
(e)	Analysing data	1
	Applying concepts	1

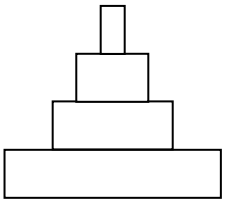
|!|BCELB0200300011|!| ★

In a class discussion, Macy and Johnny compared two types of ecological pyramids: pyramid of numbers and pyramid of biomass. They concluded that the pyramid of biomass was more accurate in reflecting the amount of energy contained within different trophic levels. The food chain and the pyramids they studied are shown below.

Food chain: A → B → C → D



Pyramid of numbers



Pyramid of biomass

Answer the following questions using the letters given in the food chain if necessary.

(a) Which organism(s) in the food chain is a / are carnivore(s)? (1 mark)

(b) Organism E feeds on the producer and is the prey of organism D.

(i) Construct a food web that includes organism E in the space below. (1 mark)

(ii) State and explain the ecological relationship between organisms B and E. (2 marks)

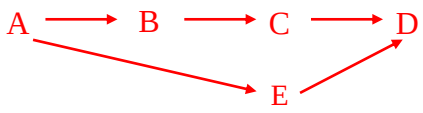
(iii) Which organism will be directly affected if herbicide is used in its habitat? (1 mark)

(c) Suggest ***two*** reasons why the pyramid of biomass is more accurate than the pyramid of numbers in reflecting the amount of energy contained within different trophic levels. (2 marks)

##

(a) C and D (1)

(b) (i)



(1)

(ii) Competition (1)

This is because both organisms B and E feed on organism A. (1)

(iii) A (1)

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(c) The pyramid of numbers fails to take into account the relative size of organisms at each trophic level. (1)

The amount of energy stored in organisms is directly proportional to the amount of living matter in their bodies (and thus, by extension, the biomass they form). (1)

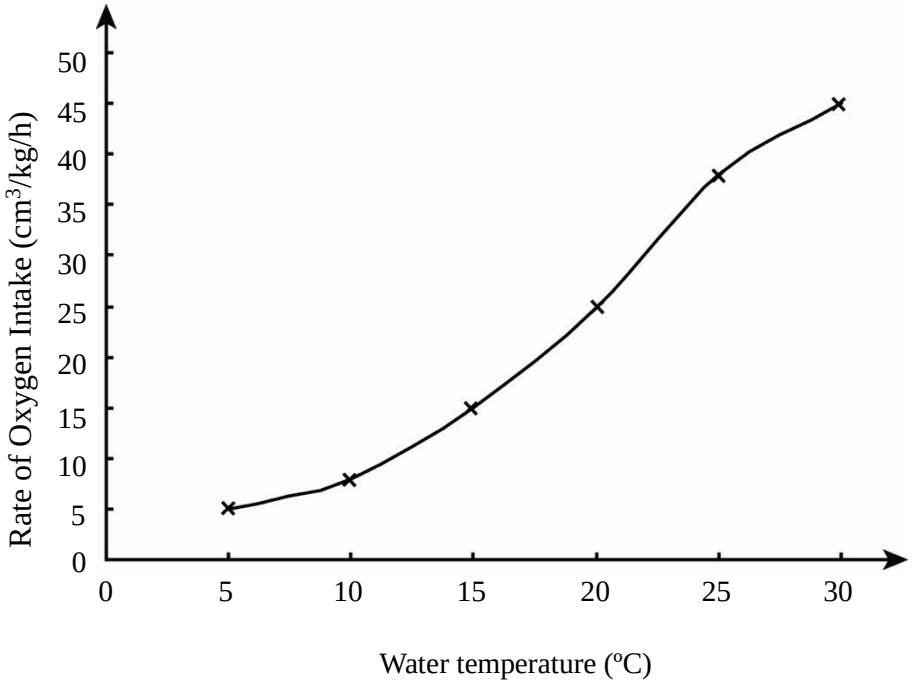
(7 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	1
(b)(i)	Applying concepts	1
(b)(ii)	Applying concepts	2
(b)(iii)	Applying concepts	1
(c)	Understanding basic concepts	2

|!|BCELA0200200012|!| ★★

During a field trip, Lily found that fishes were present in a freshwater stream. She also found that a nearby factory was discharging water at a temperature of 30 ºC into the stream. The graph below shows how the rate of oxygen uptake in the fishes changes with water temperature.



(a) Name the organ that fishes use to obtain oxygen from their surroundings. (1 mark)

(b) Calculate the volume of oxygen that would be taken up by a 2-kg fish over a 4-hour period when the water temperature is 20 ºC. Show your working. (2 marks)

(c) According to the graph, how would an increase in water temperature affect the rate of oxygen uptake in a fish? (1 mark)

(d) How would the fish population be affected after the water temperature in the stream had been raised to 30 ºC for three days? Explain your answer. (4 marks)

##

(a) Gills (1)

(b) Volume of oxygen taken up = (25 × 2 × 4) cm^3 (1)
= 200 cm^3 (1)

(c) An increase in water temperature would raise the rate of oxygen uptake in a fish. (1)

(d) The fish population would decrease. (1)
With the water temperature raised to 30 ºC, each fish must take up a higher amount of oxygen. (1)
However, the water solubility of oxygen in the stream would also be lowered. (1)
Oxygen depletion would thus occur in the stream and the fish would die from a lack of oxygen. (1)

(8 marks)

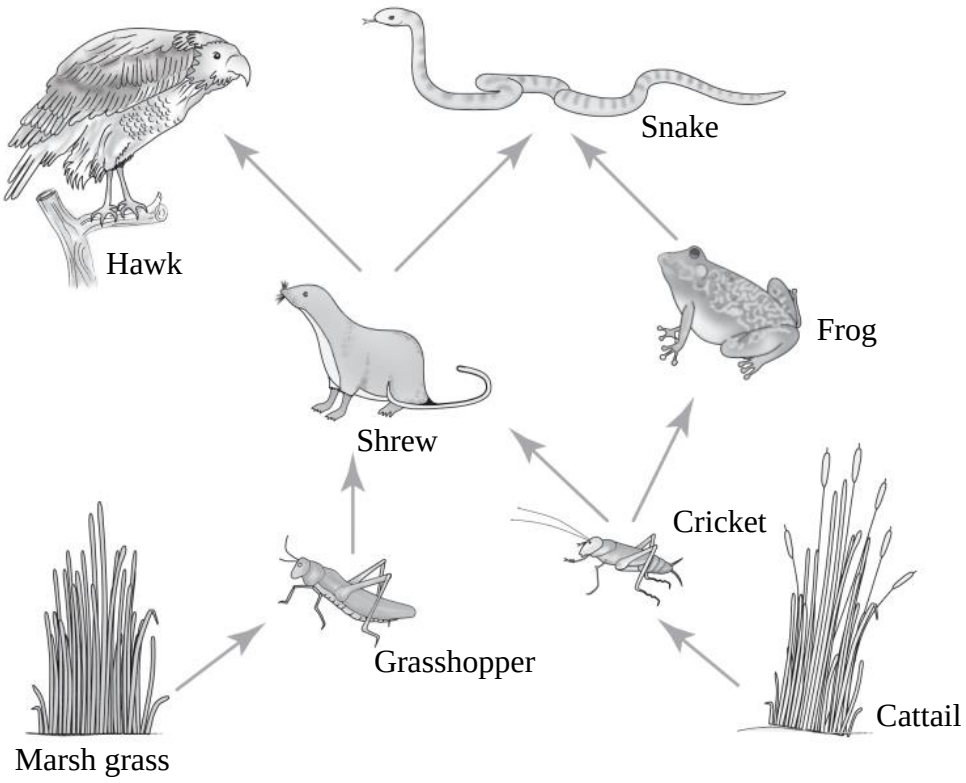
##

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	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Mathematical skills	2
(c)	Analysing data	1
(d)	Applying concepts	4

|!|BCELB0200300013|!| ★

The diagram below shows a food web of a marsh habitat.



(a) How many food chains does this food web have? (1 mark)

(b) (i) Name ***one*** organism in the food web above that is a carnivore. (1 mark)

(ii) Give ***one*** example of how this carnivore is adapted for its diet. (1 mark)

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(c) Identify ***two*** traits in grasshoppers that can prevent it from being predated on by shrews. (2 marks)

(d) (i) What is the trophic level of the snake in the food web above? (1 mark)

(ii) State ***two*** reasons why there are no organisms that prey on snakes in the habitat. (2 marks)

##

(a) 5 (1)

(b) (i) hawk / snake / shrew / frog (1)

(ii) Any ***one*** of the following: (1)

- It has sharp teeth that facilitate predation against prey.
- It can easily detect prey (e.g. hawk has good vision).

(Or other reasonable answers)

(c) Any ***two*** of the following: (2)

- The green colour of grasshoppers acts as camouflage against the marsh grass.
- Grasshoppers have wings and strong hind legs that help them escape from predators.
- Grasshoppers have antennae and compound eyes that can detect predators.

(Or other reasonable answers)

(d) (i) The 4th trophic level (1)

(ii) Any ***two*** of the following: (2)

- Due to great energy losses along the food chain, less energy is available for any organism that preys on snakes.
- Snakes have sharp fangs to provide defence against predators.
- Snakes may produce and inject venom as a means of defence.

(Or other reasonable answers)

(8 marks)

##

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	Skills assessed	Marks
(a)	Analysing data	1
(b)(i)	Analysing data	1
(b)(ii)	Applying concepts	1
(c)	Applying concepts	2
(d)(i)	Analysing data	1
(d)(ii)	Applying concepts	2

|!|BCELB0200300014|!| ★

A team of biologists visited a forest and collected data on the organisms in the habitat. The recorded figures are shown in the following table.

Organism	Density (Number of individuals / m^2)	Energy Content ($kcal/m^2/year$)
Hawk	0.08	0.004
Sparrow	120	4.2
Caterpillar	560	450
Oak tree	1.2	7000

(a) Which organism has the greatest population density? (1 mark)

(b) The following shows the food chain for the organisms above.

oak tree → caterpillar → sparrow → hawk

(i) Using the tabulated data, calculate the percentage change in energy content from oak trees to caterpillars. (Show your working and calculate to two decimal places.) (2 marks)

(ii) Give ***two*** reasons to explain the difference in energy content between oak trees and caterpillars. (2 marks)

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(c) Referring to the table, explain why the food chain above does not usually have more than four trophic levels. (2 marks)

##

(a) Caterpillar (1)

(b) (i) Percentage change = (450 – 7000) / 7000 × 100% (1)

= – 93.57% (1)

(ii) Any ***two*** of the following: (2)

- Not all oak trees are eaten by the caterpillars.
- When oak trees carry out respiration, some energy is lost as heat.

(Or other reasonable answers)

(c) There is a great loss of energy at each trophic level along the food chain. (1)

In the fourth trophic level (hawk), the amount of energy drops to 0.004 kcal/m^2/year, which may be inadequate to support organisms at higher trophic levels. (1)

(7 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)(i)	Mathematical skills	2
(b)(ii)	Applying concepts	2
(c)	Applying concepts	2

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|!|BCELB0200300015|!| ★ **SI**

In a field study of five types of grassland organisms, students recorded the amount of dry mass and the number of individuals present in a grassland habitat. The results are shown in the following table.

Organisms	Grasshopper	Hawk	Caterpillar	Alfalfa	Sparrow
Dry mass of individuals (g)	10	200	12	0.7	50
Number of individuals	50	1	41	4000	7
Total biomass (g)					

(a) (i) Complete the table above to show the total biomass for each type of organism of the study. (3 marks)

(ii) Given the calculated figures from (a)(i), construct a food web for the same grassland organisms in the space below. (2 marks)

(b) (i) Name the producer in this food web. (1 mark)

_____

(ii) What role does the producer fulfil in an ecosystem? (1 mark)

_____

_____

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(c) Categorize the grassland organisms of the study according to their modes of nutrition and diet (where applicable). (3 marks)

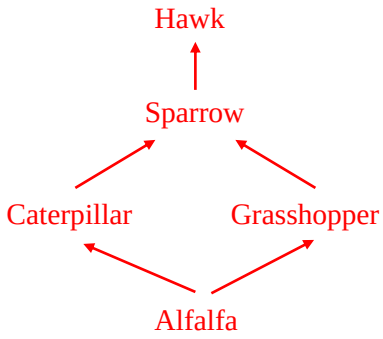
(d) If 100 grasshoppers were added into the habitat, how would the populations of the other types of organisms be affected? (2 marks)

##

(a) (i) Deduct 1 mark for each incorrect answer (Max. 3)

Organisms	Grasshopper	Hawk	Caterpillar	Alfalfa	Sparrow
Total biomass (g)	500	200	492	2800	350

(ii)



Correct order of trophic levels (1)
Use of arrows to indicate correct feeding relationship (1)

(b) (i) Alfalfa (1)
(ii) It carries out photosynthesis to synthesize organic compounds, starting the food chain. (1)

(c) The alfalfa is an autotroph. (1)
The caterpillar and grasshopper are herbivores. (1)
The sparrow and hawk are carnivores. (1)

(d) The sparrow will increase in number (1)
while the alfalfa and caterpillar will decrease in number. (1)

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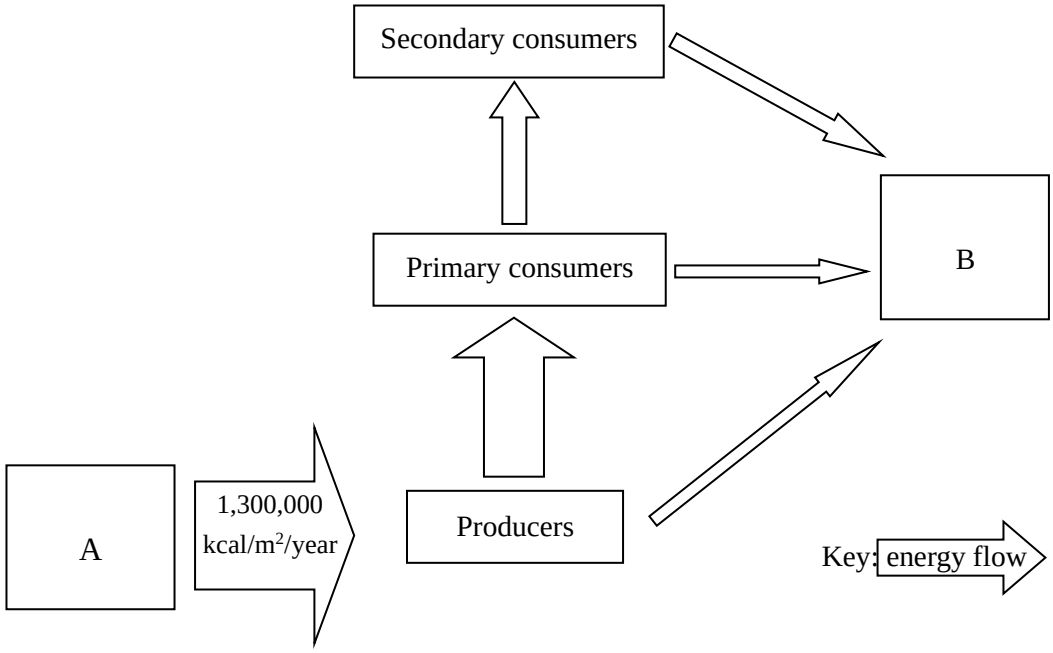
(12 marks)

##

	Skills assessed	Marks
(a)(i)	Mathematical skills	3
(a)(ii)	Presenting information	2
(b)(i)	Analysing data	1
(b)(ii)	Understanding basic concepts	1
(c)	Understanding basic concepts	3
(d)	Applying concepts	2

|!|BCELC0200300016|!|

The following diagram shows the energy flow in an ecosystem.



(a) What is A? (1 mark)

(b) Only about 1% of the energy from A is captured by producers in the food chain. Suggest ***two*** reasons why. (2 marks)

(c) The efficiency for the transfer of energy from producers to primary consumers is 13%. Calculate the total amount of energy contained within primary consumers. (1 mark)

(d) (i) Given its sources of energy, identify what type of organism B is. (1 mark)

(ii) What is the role of organism B in material cycling? (4 marks)

##

(a) The sun / solar energy / energy from the sun (1)

(b) Some wavelengths of light (e.g. green light) are not absorbed by chlorophyll for photosynthesis. (1)

Some of the sunlight is reflected when it reaches the leaf surface. (1)

(c) 1,300,000 kcal/m^2/year × 1% × 13% = 1690 kcal/m^2/year (1)

(d) (i) Decomposers (1)

(ii) Organism B breaks down the organic compounds in the excretory wastes and dead remains of producers and consumers. (1)

This process releases carbon dioxide and inorganic nutrients. (1)

Producers take up the carbon dioxide to carry out photosynthesis and synthesize food. (1)

They also absorb inorganic nutrients for growth. (1)

(9 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Applying concepts	2
(c)	Mathematical skills	1
(d)(i)	Analysing data	1
(d)(ii)	Understanding basic concepts	4

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|!|BCELB0200300017|!| ★

A scientist visited a habitat to investigate the feeding relationships of the organisms there. He constructed the following food chains to show his conclusions.

Food chain 1: oak → wood-boring beetle → blackbird → owl Food chain 2: oak → squirrel → fox

(a) What is the ecological relationship between the wood-boring beetle and the blackbird? (1 mark)

(b) Describe briefly how the owl in the food chain obtains energy from the Sun. (2 marks)

(c) Explain why the wood-boring beetles and squirrels do not exhibit competition for food. (2 marks)

(d) State ***three*** functions that oak trees perform for an ecosystem. (3 marks)

##

(a) Predation (1)

(b) Light energy from the Sun is converted into chemical energy by the oak tree during photosynthesis. (1)

The oak tree is preyed upon by the wood-boring beetle, which is in turn preyed upon by the blackbird. Eventually the blackbird is preyed upon by the owl, which obtains the chemical energy derived from the oak tree. (1)

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(c) This is because wood-boring beetles consume the wood of oak trees (1)
while squirrels consume the nuts / fruits of oak trees. (1)

(d) Any ***three*** of the following: (3)

- It provides a basic food source for other animals.
- It initiates the energy flow through the trophic levels.
- It provides shelter for other organisms.
- It maintains the balance of oxygen and carbon dioxide in the atmosphere.
- It maintains the cycling of carbon and nitrogen.

(Or other reasonable answers)

(8 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	1
(b)	Understanding basic concepts	2
(c)	Applying concepts	2
(d)	Understanding basic concepts	3

|!|BCELA0200300018|!| ★★ **SI**

In an experiment, four species (A, B, C, and D) were separated into different groups and put into three glass bottles covered with lids. The following table shows the contents of each bottle at the beginning of the experiment and the results observed after a few days.

Bottle	Contents at the beginning of the experiment	Results after a few days
1	Animal (species A)	Living
	Animal (species C)	Disappeared
	Plant (species D)	Unchanged
2	Animal (species B)	Dead
	Animal (species C)	Living
	Plant (species D)	Partially eaten
3	Animal (species A)	Disappeared
	Animal (species B)	Living
	Plant (species D)	Unchanged

(a) Explain the disappearance of species C in Bottle 1. (1 mark)

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(b) With reference to the results from Bottle 2 and 3, explain the death of species B in Bottle 2. (2 marks)

(c) According to the results, construct a possible food chain for species A, B, C and D. (2 marks)

(d) Why would it be impossible to draw any appropriate conclusions from the experiment if all the organisms were put into the same bottle? (2 marks)

(e) The relationships between another species (species X) and species A, B, C and D were studied in a similar experiment. The results are shown in the table below.

Bottle	Contents at the beginning of the experiment	Results after a few days
4	Animal (species X)	Living
	Animal (species C)	Disappeared
	Plant (species D)	Unchanged
5	Animal (species X)	Dead
	Animal (species A)	Dead
	Plant (species D)	Unchanged
6	Animal (species X)	Disappeared
	Animal (species B)	Living
	Plant (species D)	Unchanged
7	Animal (species A)	Living
	Animal (species C)	Disappeared
	Animal (species X)	Living

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According to the tables above, construct a food web to show the relationship between species X, A, B, C and D. (1 mark)

(f) If species A, B, C, D and X were present in a natural habitat and a large number of individuals of species B was subsequently introduced into the habitat, how would the other species be immediately affected? (3 marks)

##

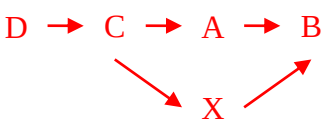
(a) Species C disappeared in Bottle 1 because it was preyed upon by species A. (1)

(b) Species A disappeared in Bottle 3 because it was preyed upon by species B. (1)
Because species A was absent in Bottle 2, species B died from a lack of food. (1)

(c) D → C → A → B
Correct order of trophic levels (1)
Use of arrows to indicate correct feeding relationship (1)

(d) If all organisms were put into the same bottle, all individuals of species D, C and A would be consumed by their respective predators, leaving only species B in the bottle. (1)
There will be insufficient data to draw definite conclusions from, making it impossible to determine the precise feeding relationship between the species. (1)

(e)



(1)

(f) The number of individuals of species A and X would decrease. (1)
The number of individuals of species C would increase (1)
while the number of individuals of species D would decrease. (1)

(11 marks)

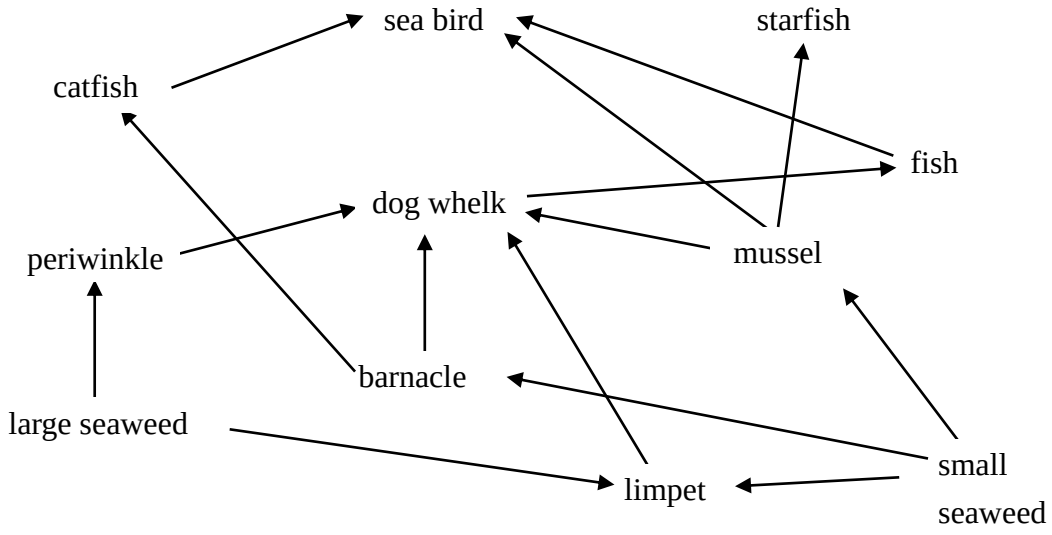
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##

	Skills assessed	Marks
(a)	Applying concepts	1
(b)	Applying concepts	2
(c)	Presenting information	2
(d)	Applying concepts	2
(e)	Presenting information	1
(f)	Applying concepts	3

|!|BCELC0200300019|!|

The figure below shows a food web for a seashore community.



(a) What is the maximum number of trophic levels in this food web? (1 mark)

(b) State the ecological relationship between sea birds and dog whelks. (1 mark)

(c) State ***two*** common needs of barnacles and limpets. (2 marks)

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(d) State ***two*** ways in which energy may be lost as it passes along a food chain. (2 marks)

(e) It is found that some barnacles grow on the back of crabs. State and explain the ecological relationship between the barnacles and the crabs. (3 marks)

##

(a) Five (1)

(b) Competition (1)

(c) Any ***two*** of the following: (2)

- Food
- Living space
- Water

(Or other reasonable answers)

(d) Any ***two*** of the following: (2)

- Most of the energy released during respiration is lost as heat to the environment.
- Some energy is lost in excretory products and faeces.
- Some organisms are not eaten by consumers and thus energy is lost.

(Or other reasonable answers)

(e) Commensalism (1)

Barnacles can benefit from the transportation and food remains provided by the crabs. (1)

The crabs are unaffected by the presence of barnacles. (1)

(9 marks)

##

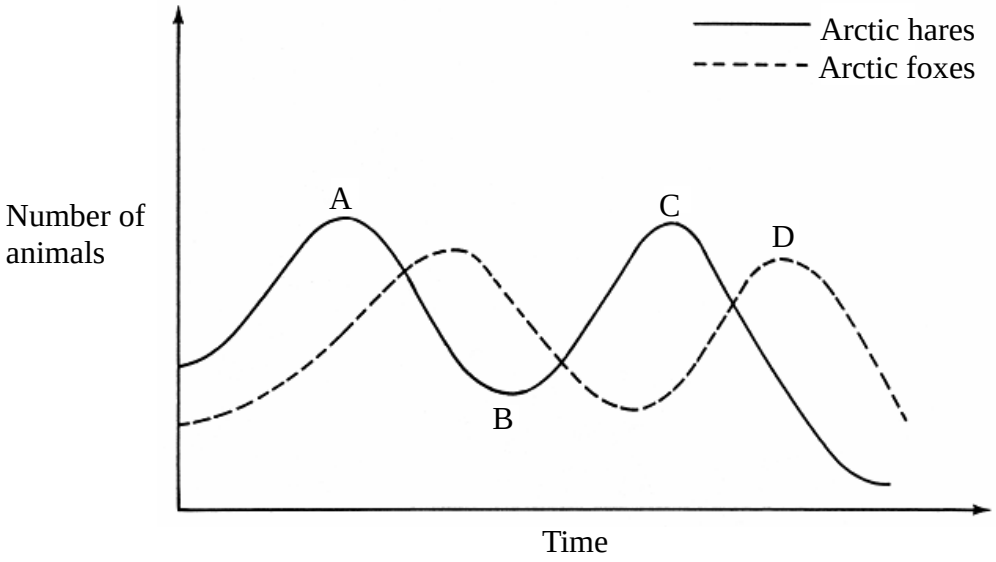
	Skills assessed	Marks
(a)	Analysing data	1
(b)	Analysing data	1
(c)	Understanding basic concepts	2

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(d)	Understanding basic concepts	2
(e)	Applying concepts	3

|!|BCELB0200200020|!| ★

The graph below shows the changes in the populations of Arctic hares and Arctic foxes over a period of time.



(a) Explain why the population of Arctic hares decreased from A to B. (2 marks)

(b) Explain why the population of Arctic hares reached a peak at C. (2 marks)

(c) Explain why the population of Arctic foxes reached a peak at D. (2 marks)

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(d) How would the population of Arctic hares be affected if most of the Arctic foxes were deliberately killed by men? Explain your answer. (4 marks)

##

(a) Arctic foxes prey on Arctic hares. (1)
Arctic foxes increased in number between A and B, thus more Arctic hares were eaten. (1)

(b) The number of Arctic foxes was low, so fewer Arctic hares were eaten. (1)
The birth rate overtook the death rate, thus the number of Arctic hares reached a peak at C. (1)

(c) More food was available to the Arctic foxes because the population of Arctic hares reached a peak at C. (1)
The abundance of prey allowed the Arctic foxes to reproduce rapidly and its population reached a peak at D. (1)

(d) Initially, the population of Arctic hares would increase rapidly (1)
because most of its predators have been killed. (1)
The larger population would eventually reduce their food supply, so competition intensifies. (1)
Many of the Arctic hares would die and their population would plateau. (1)

(10 marks)

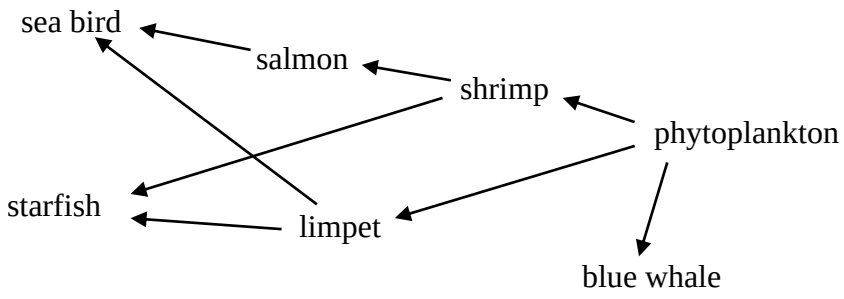
##

	Skills assessed	Marks
(a)	Applying concepts	2
(b)	Applying concepts	2
(c)	Applying concepts	2
(d)	Applying concepts	4

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|!|BCELB0200300021|!| ★

The diagram below shows a food web for a marine community.



(a) State the ecological relationship(s) between the following pairs of organisms.

(i) sea bird and starfish (1 mark)

(ii) shrimp and starfish (1 mark)

(b) (i) State the role fulfilled by phytoplankton in the food web. (1 mark)

(ii) Give ***one*** abiotic factor that may affect the number of phytoplankton. (1 mark)

(c) Although blue whales consume only phytoplankton, they weigh 120 tons on average and are the largest mammals on Earth.
With reference to their diet, explain why blue whales can grow to enormous sizes. (3 marks)

(d) Describe and explain what would happen to the number of salmon if all the starfish in the community had been killed. (3 marks)

##

(a) (i) Competition (1)

(ii) Predation (1)

(b) (i) It acts as the producer that provides food for the consumers. (1)

(ii) Temperature / light intensity / carbon dioxide concentration / other reasonable answers (1)

(c) The food chain is short and consists of only two trophic levels: phytoplankton and blue whale. (1)

Thus the energy loss incurred from the transfer of energy between trophic levels is minimized. (1)

Blue whales also consume an exceptionally large amount of phytoplankton every day, which greatly increases their biomass. (1)

(d) Shrimp would no longer be consumed by starfish. (1)

More shrimp would then be available to the salmon as food. (1)

The salmon would produce more offspring and their numbers would increase. (1)

(10 marks)

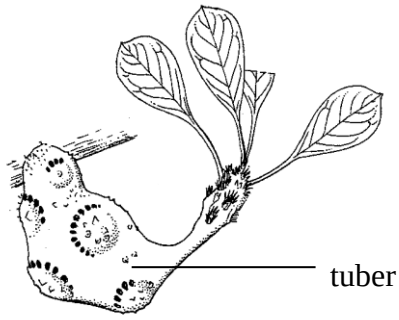
##

	Skills assessed	Marks
(a)(i)	Analysing data	1
(a)(ii)	Analysing data	1
(b)(i)	Analysing data	1
(b)(ii)	Understanding basic concepts	1
(c)	Applying concepts	3
(d)	Applying concepts	3

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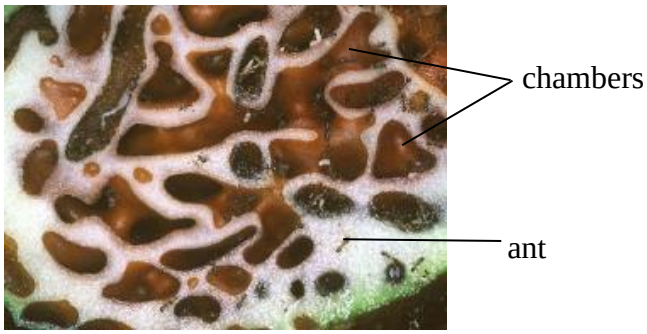
|!|BCELA0200200022|!| ★★ **Cross topic**

The diagram below shows an ant-house plant found in a rainforest. Ant-house plants have enlarged stems that form tubers with many air-filled chambers within which ants may live. The ants excrete their organic wastes within the chambers and aggressively attack other insects on the plant.



(a) With reference to the diagram above, to which group of flowering plants does the ant-house plant belong? Explain your answer. (2 marks)

(b) The photograph below shows the transverse section of a tuber of the ant-house plant.



Suggest ***one*** advantage to having so many air-filled chambers within the tuber. (1 mark)

(c) Explain how the ant-house plant can gain extra nutrients from the ants. (2 marks)

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(d) What might happen to the ant-house plant if all the ants were removed? Explain your answer. (2 marks)

(e) State ***two*** benefits that the ants gain from the ant-house plant. (2 marks)

(f) State the ecological relationship between the ant and the ant-house plant. (1 mark)

##

(a) Dicotyledonous plant (1)
This is because the plant possesses leaves with a branching network of veins. (1)
(b) The tuber can carry more ants. (1)
(c) The organic wastes excreted by the ants are broken down into inorganic nutrients by decomposers. (1)
The inorganic nutrients can then be readily absorbed by the chamber surfaces. (1)
(d) The ants attack other insects on the plant, thus providing protection to its host. (1)
If the ants were removed, the plant would lose this protection and may be exposed to attacks by other insects. (1)
(e) Any ***two*** of the following: (2)
- The ant-house plant provides a habitat for the ants.
- The ant-house plant can protect the ant from predators.

(Or other reasonable answers)
(f) Mutualism (1)

(10 marks)

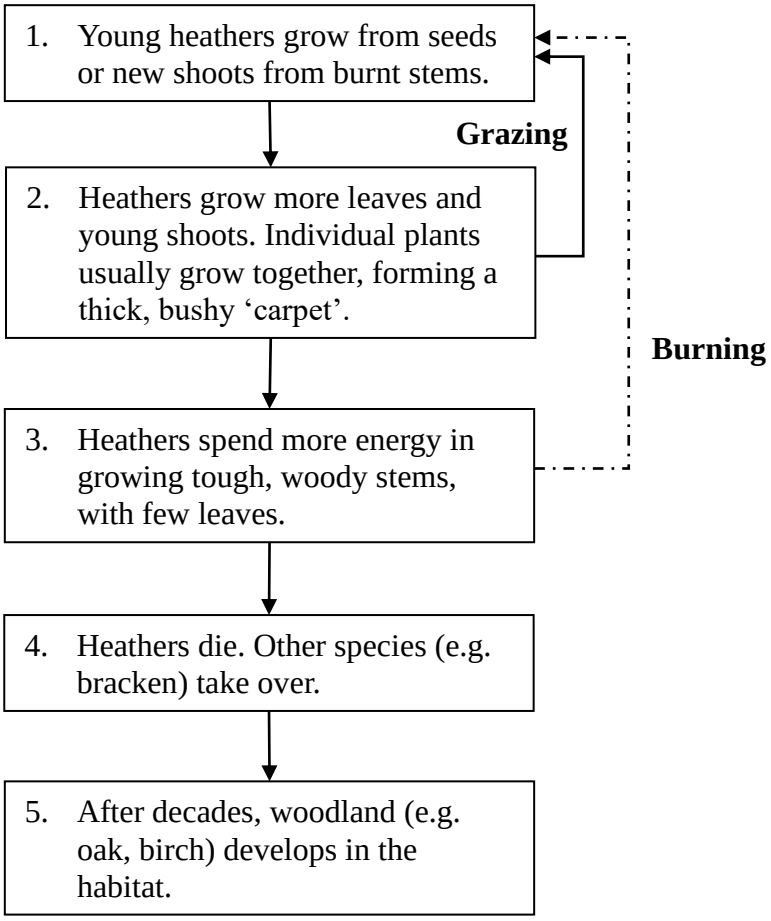
##

	Skills assessed	Marks
(a)	Analysing data	1
	Understanding basic concepts	1
(b)	Applying concepts	1
(c)	Applying concepts	2
(d)	Applying concepts	2
(e)	Applying concepts	2
(f)	Applying concepts	1

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|!|BCELA0200200023|!| ★★ **Cross topic**

The following diagram shows the changes in the vegetation of a region where succession occurs.



(a) What biological term can describe the mature woodland established at the final stage of the succession above? (1 mark)

_____

(b) If the region is used for cow grazing, the heathers will be burnt every few years to encourage the growth of new plants. Explain how such burning may increase the food available to the cows. (2 marks)

_____

_____

_____

_____

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(c) State ***one*** potential threat that the burning of vegetation would pose to the environment. (1 mark)

(d) (i) Name the process that is exemplified by the regrowth of heathers in the region after it was burnt. (1 mark)

(ii) Give ***three*** reasons why a community can be established faster on burnt lands that previously had existing communities compared to bare lands. (3 marks)

##

(a) Climax community (1)

(b) Heathers grow and gradually develop tough, woody stems which are unedible to cows. (1)

After burning, more edible leaves and shoots will grow. (1)

(c) Any ***one*** of the following: (1)

- Such burning emits carbon dioxide that enhances global warming.
- Such burning releases soot particles that cause air pollution.
- The soil, no longer covered by vegetation, becomes susceptible to soil erosion.

(Or other reasonable answers)

(d) (i) Secondary succession (1)

(ii) Any ***three*** of the following: (3)

- Fertile soil is already present / Ash provides nutrients in soil.
- The soil contains seeds and spores, which can grow into new plants.
- Plant roots are undamaged by the burning and can grow into new plants.
- Burrowing animals survive.

(Or other reasonable answers)

(8 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Applying concepts	2
(c)	Understanding basic concepts	1
(d)(i)	Understanding basic concepts	1
(d)(ii)	Understanding basic concepts	3

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|!|BCELC0200800024|!| **SI**

A field study was conducted to find the distribution of various plant species on a beach. A transect was placed on the beach with ten different quadrats placed at different positions along the transect. A soil sample was taken within each quadrat to measure the mass of organic materials present. The results of the study are shown in the following table.

Quadrat number	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
Distance from top of the beach (m)	0	80	170	250	500	650	900	1600	2000	2400
Number of plant species found	2	2	6	7	20	6	5	3	4	2
Total number of plants found	30	40	43	60	98	72	34	40	59	30
Mass of organic materials (g)	0.3	0.4	0.2	0.3	2	7.6	8.8	9.3	10.2	10.5

(a) Explain why a transect is more suitable than random sampling in this study. (2 marks)

(b) In the study, a transect was used and quadrats were placed at different positions along the transect. What is the name of this sampling method? (1 mark)

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(c) Compare the data collected from quadrat 1with those from quadrat 5. (3 marks)

(d) Give an advantage and disadvantage to using quadrats for sampling. (2 marks)

(e) Give ***one*** reason why plants can survive under the dry conditions on the beach. (1 mark)

##

(a) Sampling with transects are appropriate for habitats where there are gradual variations in population and environmental conditions. (1)
Random sampling does not show the range of species distribution. (1)

(b) Belt transect (1)

(c) More species were found in quadrat 5. (1)
Plant coverage was greater in quadrat 5. (1)
There was more organic matter in quadrat 5. (1)

(d) Advantage: easy to conduct (1)
Any ***one*** of the following disadvantages: (1)
- hard to count fast-moving animals
- limited to vegetation and slow-moving or stationary animals
- the quadrat must be placed on fairly flat land

(e) Any ***one*** of the following: (1)
- Their leaves are covered with a thick cuticle.
- Their leaves are small or needle-shaped.

(Or other reasonable answers)

(9 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	1
(c)	Analysing data	3
(d)	Understanding basic concepts	2
(e)	Understanding basic concepts	1

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